

University of Nairobi (UoN) Automated Missing Marks & Academic Grievance Clearinghouse

Web Application Development Specification

Prepared for: Web Development Contractor / Engineering Team

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Document Status: Draft for Developer Scoping & Quotation

1. Executive Summary

The University of Nairobi (UoN) requires a purpose-built, web-based **Missing Marks & Academic Grievance Clearinghouse** to replace ad-hoc, paper-based, and email-driven processes for resolving missing, disputed, or incomplete student marks. The system must provide a structured, auditable, multi-tier ticketing pipeline connecting **students, lecturers, and Heads of Department (HODs) / Exam Officers / Registrar staff**, across all 10 UoN Faculties, with built-in bi-directional communication, proof-of-claim document handling, and status tracking through to Student Management System (SMS) reconciliation.

This document specifies functional requirements, user roles, data validation rules, workflow states, technical architecture options, and security constraints sufficient for a development team to scope, estimate, and build the system.

2. Target Institution & Scope

The system must operate across **all 10 UoN Faculties**:

#	Faculty
1	Faculty of Science & Technology
2	Faculty of Health Sciences
3	Faculty of Engineering
4	Faculty of Business & Management Sciences
5	Faculty of Arts & Social Sciences
6	Faculty of Law
7	Faculty of Education
8	Faculty of Built Environment & Design
9	Faculty of Agriculture
10	Faculty of Veterinary Medicine

Each faculty contains multiple departments and unit codes, which must be configurable/administrable rather than hardcoded, to accommodate curriculum changes.

3. System Architecture Overview

The application is a **multi-portal, role-gated web system** with three authenticated tiers sitting behind a public entry gateway. All tiers operate against a shared ticket data model but present role-specific views and permissions.

```
Step 0: Dynamic Faculty Gateway (public, unauthenticated)
|
├─ Tier 1: Student Intake & Tracking Portal (@student.uonbi.ac.ke SSO)
├─ Tier 2: Lecturer Portal (@uonbi.ac.ke SSO)
└─ Tier 3: HOD / Exam Officer / Registrar Portal (@uonbi.ac.ke SSO, elevated role)
```

All three tiers read/write to a single **Ticket** entity with a shared status lifecycle, but access is scoped by faculty, department, unit code, and role.

4. Step 0 — Dynamic Faculty Gateway

- Public-facing landing page, no authentication required to *view* (authentication required to *submit*).
- Faculty selector (dropdown or card grid) filters a dependent Department selector.
- Department selection filters an optional Unit Code selector (auto-populated from a course catalog table).
- Gateway routes the user into the correct SSO flow based on their subsequent role selection (Student vs. Staff).

Data dependency: Requires a maintained **Faculty → Department → Unit Code** reference table, ideally admin-editable via Tier 3 without a code deployment.

5. Tier 1 — Student Intake & Tracking Portal

Authentication: Google SSO restricted to `@student.uonbi.ac.ke` domain only. Any other domain must be rejected at the OAuth consent/callback stage.

5.1 Ticket Submission Form — Field Specification

Field	Type	Validation Rule	Required
Faculty	Dropdown	Must match Step 0 selection	Yes
Department	Dropdown	Dependent on Faculty	Yes
Registration Number	Text	Regex: <code>^([A-Z0-9]{2,3})\(([0-9]{5,6})\)/([0-9]{4})\$</code>	Yes
Unit Code	Dropdown/Text	Must exist in course catalog for selected department	Yes
Assessment Category	Dropdown	One of: End of Semester Main Exam, CAT, Lab Report/Practical Score, Fieldwork/Attachment Grade, Research Project/Dissertation Grade	Yes
Claimed Score / Detail	Numeric or Text	Range-checked against assessment category's max marks (e.g., CAT ≤ 30)	Yes
Proof Attachment(s)	File Upload	PDF or JPEG only; max 5MB per file; multiple files allowed	Yes (min. 1 file)
Authorization Disclaimer	Checkbox	Must be checked to enable submit button	Yes
Additional Notes	Text area	Free text, max 1000 characters	No

5.2 Authorization Disclaimer

By submitting this claim, I affirm that the information and documentation provided are true, accurate, and represent my own academic work. I understand that submission of falsified proof or fraudulent claims constitutes academic misconduct under University of Nairobi regulations and may result in disciplinary action, including but not limited to suspension or expulsion. I consent to my claim being reviewed by the relevant course lecturer, Head of Department, and Examination Officer as part of the standard grievance resolution process.

5.3 Student Tracking Dashboard

- List/table of the student's own submitted tickets with: Ticket ID, Unit Code, Assessment Category, Status, Last Updated timestamp.
- Click-through to a **Ticket Detail & Thread View** showing:
 - Original submission details and uploaded proof (view-only, not re-editable once submitted).
 - Full chronological feedback thread (lecturer comments, student replies).
 - Ability to post replies and upload **supplementary** proof files without opening a new ticket.
- Status badge is read-only for students; only the Tier 2/3 roles can change ticket status.

6. Tier 2 — Dedicated Lecturer Portal & Tab

Authentication: Google SSO restricted to `@uonbi.ac.ke` staff domain only, cross-checked against a lecturer-to-unit-code assignment table so lecturers see only claims for units they teach.

6.1 Core Features

- Unit Code Summary Cards:** One card per assigned unit code, showing total flagged/open claims, broken down by assessment type.
- Claims Filter Bar:** Filter by assessment type (Main Exam / CAT / Lab-Practical / Fieldwork / Research Project), by status, and by submission date range.
- In-Browser Proof Viewer:** Modal viewer supporting inline PDF/JPEG rendering of exam slips, CAT dockets, or lab attendance/signature sheets — no forced download required.
- Status & Score Action Panel** per ticket:

Lecturer Action	Effect	Required Input
Verified — Forwarded to HOD	Ticket status → "Under Departmental Processing"; score is recorded	Numeric score entry
Rejected — Insufficient Proof	Ticket status → "Rejected"; student notified	Mandatory explanation note (min. 20 characters)
Request More Information	Ticket remains open; flagged "Awaiting Student Response"	Free-text feedback message

6.2 Bi-Directional Feedback & Commenting Engine

- **Lecturer Feedback Module:** Lecturers post timestamped comments or information requests directly on a ticket thread (e.g., asking for a clearer scan of a specific page).
- **Student Response Interface:** Students reply and attach supplementary files within the same thread — no duplicate ticket creation.
- **Real-Time Email Alerts:**
 - Lecturer comment → instant email to student's `@student.uonbi.ac.ke` address.
 - Student reply → instant in-portal notification (and optionally email) to the lecturer.
- All thread entries are immutable once posted (append-only audit trail); edits are not permitted, only new replies.

7. Tier 3 — HOD, Exam Officer & Registrar Master Portal

Authentication: Google SSO on `@uonbi.ac.ke` domain with an elevated role flag (HOD / Exam Officer / Registrar), assignable by a system super-admin.

7.1 Status Lifecycle (Full Pipeline)

Status	Meaning	Who Can Set It
Submitted to Department/Lecturer	Initial state on student submission	System (automatic)
Under Departmental Processing	Lecturer has verified and forwarded with a score	Lecturer (Tier 2)
Rejected — Insufficient Proof	Lecturer declined the claim	Lecturer (Tier 2)
Awaiting Student Response	Lecturer requested more info	Lecturer (Tier 2)
Cleared for SMS Update	HOD/Exam Officer has approved the verified score for entry into SMS	HOD/Exam Officer (Tier 3)
Verified on SMS	Registrar confirms the mark is reflected in the official Student Management System	Registrar (Tier 3)
Closed — Escalated	Manually closed for administrative/disciplinary escalation outside the normal flow	Registrar (Tier 3)

7.2 Administrative Features

- **Master Dashboard:** Real-time counts of tickets by status, faculty, department, and assessment type; aging report (days-in-status) to flag stalled tickets.
- **Visual Analytics:** Charts for volume trends, resolution time distribution, and rejection-rate by department/lecturer (for quality oversight).
- **One-Click CSV/Excel Master Export:** Full export of ticket data filtered by faculty/department/date range, formatted for Faculty Board clearance packets (columns: Reg No., Unit Code, Assessment Type, Claimed Score, Verified Score, Status, Lecturer, Dates).
- **Reference Data Administration:** CRUD interface for Faculty → Department → Unit Code catalog, and for lecturer-to-unit-code assignments (avoids requiring developer intervention for routine curriculum changes).

8. Technical Architecture & Constraints

8.1 Stack Options

Option	Stack	Trade-offs
A — Full Web App	Next.js + Tailwind CSS (frontend), Node.js API layer, PostgreSQL via Supabase	Higher control, scalability, proper relational integrity, file storage via Supabase Storage; requires hosting cost and DevOps ownership
B — Zero-Cost Google Workspace	Google Apps Script + Google Sheets as datastore, Google Drive for file storage	No hosting cost, native Google SSO integration, faster to deploy; weaker at scale, limited concurrent-write integrity, harder to enforce complex RegEx/relational validation

The contractor should propose one option with justification, or a phased approach (Option B as MVP, migrating to Option A at scale).

8.2 Authentication — Dual-Domain OAuth

- Strict domain separation is **mandatory**:
 - `@student.uonbi.ac.ke` → Student Portal only.
 - `@uonbi.ac.ke` → Lecturer/Admin Portals only, gated further by role assignment (Lecturer vs. HOD/Exam Officer/Registrar).
- Cross-domain login attempts (e.g., a staff account attempting the student portal) must be rejected with a clear error, not silently downgraded.

8.3 File Pipeline & Security

- All uploads are automatically **sanitized and renamed** server-side using the pattern:

```
[TICKET_ID]_[REG_NUMBER]_[FILE_TYPE].[ext]
```

- Accepted formats: PDF, JPEG only. Reject all other MIME types at both client and server validation layers (never trust client-side extension checks alone).
- Max file size: 5MB per file, enforced server-side.
- Files should be scanned/sanitized against embedded scripts/macros (particularly relevant for PDFs) before storage.
- Access to uploaded files must be permission-scoped (only the submitting student, assigned lecturer, and Tier 3 staff for that faculty/department can retrieve a given file) — no public or guessable direct file URLs.

8.4 Data Minimization

The system must **not** collect:

- National ID numbers
- Financial details (bank accounts, M-Pesa numbers, etc.)
- Passwords (authentication is SSO-only; no local password storage)
- Health records

Only academic data (registration number, unit code, assessment scores, proof-of-claim documents) and institutional email identity should be stored.

9. Non-Functional Requirements

Category	Requirement
Audit Trail	Every status change and comment must be immutably logged with actor identity and timestamp
Notifications	Email alerts must be near-real-time (target: under 2 minutes from trigger event)
Availability	Target uptime aligned with academic calendar peak periods (results release, clearance windows)
Accessibility	Interface should meet basic WCAG 2.1 AA considerations given mixed-device student access
Data Retention	Define retention policy for resolved tickets (recommend minimum 1 academic year, subject to UoN records policy)
Localization	English-language interface; Kenyan date/number formatting conventions

10. Developer Deliverables

1. **Full source code repository** (frontend + backend/scripts) with README and setup instructions.
2. **Deployed working system** on an agreed environment (staging + production), with the chosen stack (Option A or B) clearly documented.
3. **Database/Sheets schema documentation**, including the Faculty/Department/Unit Code reference model and Ticket entity model.
4. **Admin guide** for managing the reference data (faculties, departments, unit codes, lecturer assignments) without developer involvement.
5. **User guides** (short, role-specific) for Student, Lecturer, and HOD/Exam Officer/Registrar portals.
6. **Test evidence**: validation testing for the registration-number RegEx, file-upload constraints, dual-domain OAuth enforcement, and status-transition rules.
7. **Security summary**: confirmation of data-minimization compliance, file-access scoping, and sanitization pipeline.
8. **Post-launch support terms** (bug-fix window, SLA for critical issues).

11. Financial Quotation Request

Developers/vendors responding to this brief should submit a quotation covering:

- **Development cost breakdown** by module (Step 0 Gateway, Tier 1, Tier 2, Tier 3, Notification Engine, Admin/Reference Data tooling).
- **Cost comparison** between Stack Option A (Next.js/Supabase) and Stack Option B (Google Workspace/Apps Script), including any recurring hosting/storage costs for Option A.
- **Timeline** with milestones (design sign-off, MVP, UAT, production launch).
- **Post-launch maintenance and support pricing** (monthly/annual retainer or per-incident).
- **Any assumptions or exclusions** (e.g., SMS system integration depth, number of concurrent users supported, data migration from legacy processes).

Quotations should be itemized rather than a single lump sum, to allow the Office of Academic Affairs to evaluate cost-effectiveness per module against the described scope.

